

1a	The <u>net momentum of a system remains constant</u> provided <u>no external resultant force</u> acts on the system.	B1 B1
1bi	(total initial momentum in the direction at right-angles to the direction of the initial path of ball A = 0) (taking $\uparrow$ as positive) $0 = 4.0 \times 6.0 \sin \theta - 12 \times 3.5 \sin 30^\circ$ $\theta = 61^\circ$	C1 A1
1bii	Considering momentum along the direction of the initial path of ball A: By Conservation of linear momentum, total initial momentum = total final momentum $4.0 \times v = 4.0 \times 6.0 \cos 61^\circ + 12 \times 3.5 \cos 30^\circ$ $v = 12 \text{ m s}^{-1}$	C1 A1
1biii	total initial k.e. = $\frac{1}{2} (4)(12)^2 = 288 \text{ J}$ total final k.e. = $\frac{1}{2} (4)(6.0)^2 + \frac{1}{2} (12)(3.5)^2 = 145.5 \text{ J}$ Since the total kinetic energy before and after collision is different, the collision is inelastic.	M1 A1

2ai	take any two sets of coordinates to determine a constant value (E/x)	M1
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